

Dhruv Nirmal

Melbourne, Australia | dhruvnirmal2111@gmail.com | +61 406 259 619 | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

I build machine learning and predictive models that turn large, messy datasets into decisions a business can actually act on, and then make sure the finding lands with the people who have to act on it. At a data and analytics consulting firm, I led a classification model over roughly **A\$12B** of client spend and built a pricing analysis that helped a client find around **A\$2M** in savings in one product category, working directly with commercial stakeholders from category managers up to CFOs. I like owning a modelling problem end to end: framing it, engineering the features, tuning and validating the model, then explaining what it means in plain language through a dashboard or a conversation. AGL's Data Science Chapter, sitting across Acquisition Performance and Enablement and Product and Pricing Analytics, is exactly the kind of commercial, model-to-decision work I want to keep doing.

KEY SKILLS

Machine Learning and Predictive Modelling: scikit-learn, Classification (TF-IDF/NLP), Logistic Regression, Time-Series Forecasting (Prophet), Anomaly Detection, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Class-Imbalance Handling

Programming and Data: Python, SQL, PySpark, Databricks, Snowflake, Large and Complex Datasets

Visualisation and Storytelling: Tableau (16+ dashboards), Power BI, Data Storytelling, KPI Analytics, Executive Dashboards

Commercial and Pricing Analytics: Pricing Strategy, Cost and Spend Analysis, Stakeholder Communication, Experimentation and Hypothesis-Driven Analysis

AI and Innovation: Claude, ChatGPT, Agentic AI Workflows, Multi-Agent Orchestration, NLP/Sentiment Analysis, Prompt Engineering

EXPERIENCE

Data Scientist, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 to Present | Melbourne, Australia

A data and analytics consulting firm. I build and deploy machine learning, forecasting and anomaly-detection models on large-scale client datasets, then work directly with client stakeholders, from category managers to CFOs, to turn what the models find into pricing, sourcing and commercial decisions.

- Led the company's first ML classification project: a one-vs-all binary classifier (TF-IDF, feature engineering, class-imbalance handling, cross-validation, hyperparameter tuning) built over **~A\$12B** of client spend across five years, cutting a manual categorisation cycle from 1 to 1.5 months down to a single day's model run; the documented pipeline is now rolled out across the wider client base.
- Owned pricing and spend analytics for a restaurant chain (AU/NZ) as sole analyst: built a pricing-decision dashboard and supplier-onboarding tracker surfacing price movement and cost leakage down to product, venue and supplier level, helping the client identify roughly **A\$2M a year** (about 30%) in savings in their largest category, presented directly to the CFO, Chief Procurement Officer and category managers.
- Built a Prophet time-series forecasting model incorporating external demand drivers (sea-freight trends, input pricing) to forecast raw-material and inventory requirements three months ahead for a fresh-produce client, achieving a 12.5 to 14% error margin comfortably within the client's tolerance, delivered through a dashboard.
- Ran anomaly detection across procurement datasets to surface cost-leakage patterns contributing to roughly 15% client savings, with findings delivered through 16+ Tableau dashboards used by procurement and finance teams to track KPIs and act on cost-optimisation opportunities.

- Designed a company-wide agentic AI workflow (Claude) that auto-generates structured client progress reports, adopted across the full analyst team (about 25 to 30 accounts) and saving roughly 12 to 15 hours a week, alongside a multi-agent QA system that checks model output before it reaches a client, an example of building AI into a live workflow rather than just using it.
- Built and maintain 20+ automated ETL pipelines (Python, SQL, Azure) processing millions of transactions monthly across 5 enterprise clients, including end-to-end ownership of the company's largest client account (5 global regions, roughly 6M invoice rows refreshed weekly).

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 to Nov 2023 | Melbourne, Australia

- Built a distributed anomaly-detection pipeline (Databricks, PySpark) improving detection accuracy by about 20% on large-scale procurement data, engineering scalable transformation layers for cloud processing.
- Deployed a Power BI analytics solution adopted by senior Department of Transport stakeholders, presenting model architecture and outcomes to cross-functional, non-technical audiences.

Research Data Analyst, Terminal Ballistics Research Laboratory (TBRL), DRDO

Jan 2021 to Jul 2021

- Built statistical models to predict blast-wave noise levels, improving prediction accuracy by about 20%, and compiled findings into a structured report for defence engineering teams.

PROJECTS

MyFacit: Analytics and Decision-Support SaaS for Hospitality (Forecasting, Pricing, BI)

- Built a centralised analytics platform for a small hospitality business, integrating 8+ data sources into one consistent view with dashboards, short-term forecasting and insights for pricing, product mix and operations.
- In pilot with a local business as design partner and in active use, with weekly sessions to refine it against real-world data, turning pricing and demand signals into decisions a non-technical operator can act on.

Job Hunt OS: Agentic AI Career System (Python, Claude Code, Multi-Agent)

- Designed and built a multi-agent AI system (orchestrator plus specialist agents) that tailors resumes to a job description, drafts cover letters and runs a two-track outreach process, with a human review gate at every output stage.
- Demonstrates the same discipline the JD asks for: prompt and system design, evaluating AI output critically rather than at face value, and shipping something that runs unattended in production.

EDUCATION

Master of Data Science, Monash University | Feb 2022 to Dec 2023

Coursework: Statistics I/II, Machine Learning, Applied Forecasting, High Dimensional Analysis, Communicating with Data

Bachelor of Engineering, Thapar University | May 2017 to May 2021